

Trainings ▾

Preparatory Steps

Industrial and Power Generation

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



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⚠ WARNING ⚠

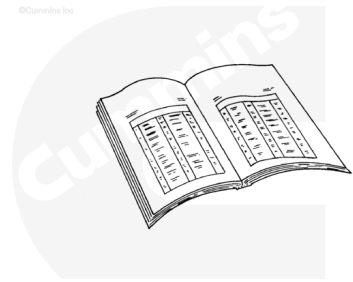
Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Drain the cooling system. Refer to Procedure 008-018 in Section 8 (</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>).
- Remove the rocker lever covers. Refer to Procedure 003-011 in Section 3 (</qs3/pubsys2/xml/en/procedures/56/56-003-011-tr.html>).
- Remove the rocker lever assembly. Refer to Procedure 003-009 in Section 3 (</qs3/pubsys2/xml/en/procedures/56/56-003-009-tr.html>).
- Remove and mark the push rods so they can be installed in their original positions. Refer to Procedure 004-014 in Section 4 (</qs3/pubsys2/xml/en/procedures/56/56-004-014-tr.html>).
- Remove the crossheads. Refer to Procedure 002-001 in Section 2 (</qs3/pubsys2/xml/en/procedures/56/56-002-001-tr.html>).

QSK60 Marine Applications

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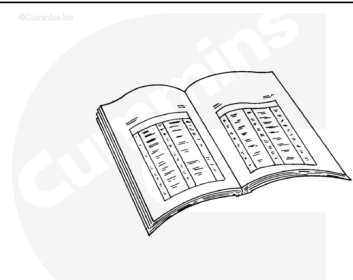
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- Drain the cooling system. Refer to Procedure 008-018 in Section 8 (</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>).
- Remove the aftercooler heat shields. Refer to Procedure 010-129 in Section 10 (</qs3/pubsys2/xml/en/procedures/56/56-010-129-tr.html>).
- Remove the rocker lever covers. Refer to Procedure 003-011 in Section 3 (</qs3/pubsys2/xml/en/procedures/56/56-003-011-tr.html>).
- Remove the rocker lever assembly. Refer to Procedure 003-009 in Section 3 (</qs3/pubsys2/xml/en/procedures/56/56-003-009-tr.html>).
- Mark the push rods so they can be installed in their original positions. Refer to Procedure 004-014 in Section 4 (</qs3/pubsys2/xml/en/procedures/56/56-004-014-tr.html>).
- Remove the crossheads. Refer to Procedure 002-001 in Section 2 (</qs3/pubsys2/xml/en/procedures/56/56-002-001-tr.html>).

with Electronically Actuated Injector

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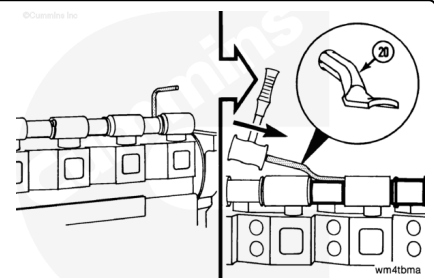
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Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

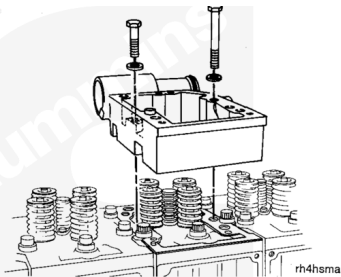
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- Remove and mark the push rods so they can be installed in their original positions. Refer to Procedure 004-014 in Section 4 (</qs3/pubsys2/xml/en/procedures/56/56-004-014-tr.html>).
- Remove the crossheads. Refer to Procedure 002-001 in Section 2 (</qs3/pubsys2/xml/en/procedures/56/56-002-001-tr.html>).
- Remove the injectors. Refer to Procedure 006-026 in Section 6 (</qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html>).

Remove

Use Water Tube Driver, ST-1319. Drive the tubes on the right bank toward the rear of the engine. Drive the tube on the left bank toward the front of the engine. Drive the tube until it clears the rocker lever housing.



Remove the seven rocker lever housing mounting capscrews and the rocker lever housing.



Remove and discard rocker lever housing gasket.



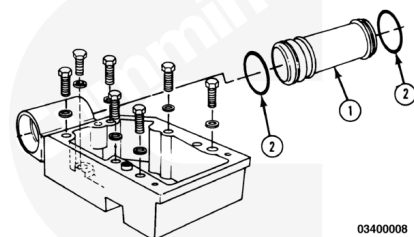
Clean and Inspect for Reuse

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.



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Remove the following parts:

1. Water transfer tube
2. O-rings.

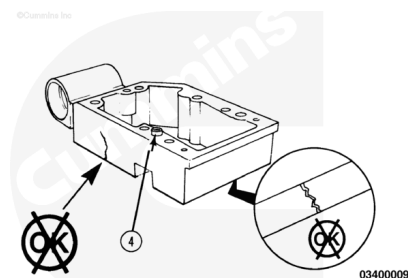
Discard the o-rings.

Use steam or solvent to clean the parts.

Inspect the parts for damage. If the ring dowel (4) is damaged, it **must** be replaced. Use a blind hole puller contained in light-duty puller kit, Part Number 3375784, or equivalent, to remove

the dowel ring.

Use the crack detection kit, Part Number 3375432, to inspect the housing for cracks. If the housing is cracked, it **must** be replaced.



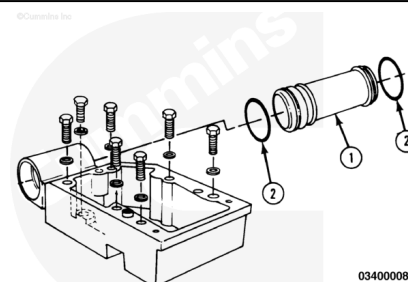
Install

Lubricate the o-rings (2) with vegetable oil.

Note : Use of other lubricants, such as engine oil, will cause the o-rings to swell.

Install the o-rings.

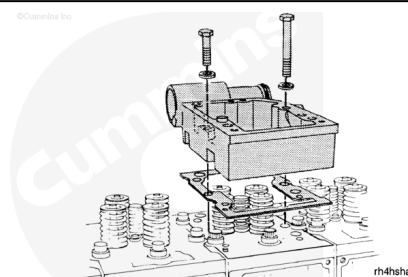
Install the water transfer tube (1).



The gasket and housing **must** be aligned correctly with the dowel pins.

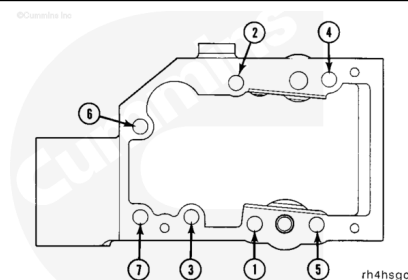
Align and install the rocker lever housing gasket and the rocker lever housing.

Install the seven rocker lever housing mounting capscrews.

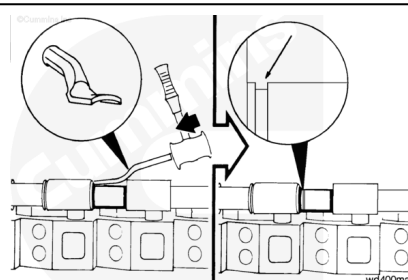


Use the sequence shown to tighten the capscrews.

Torque Value: 115 n•m [85 ft-lb]



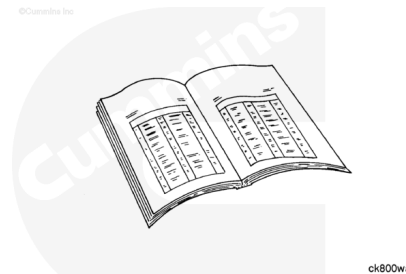
Use water tube driver, Part Number ST-1319, and a hammer. Drive the water tube into the adjacent housing until the water tube is centered.



Finishing Steps

Industrial and Power Generation

- Install the push rods. Refer to Procedure 004-014 in Section 4 (/qs3/pubsys2/xml/en/procedures/56/56-004-014-tr.html).
- Install the crossheads. Refer to Procedure 002-001 in Section 2 (/qs3/pubsys2/xml/en/procedures/56/56-002-001-tr.html).
- Install the rocker lever assemblies. Refer to Procedure 003-009 in Section 3 (/qs3/pubsys2/xml/en/procedures/56/56-003-009-tr.html).
- Adjust the overhead. Refer to Procedure 003-006 in Section 3 (/qs3/pubsys2/xml/en/procedures/56/56-003-006-tr.html).
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3 (/qs3/pubsys2/xml/en/procedures/56/56-003-011-tr.html).
- Fill the cooling system. Refer to Procedure 008-018 in Section 8 (/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html).
- Operate the engine to 71°C [160°F] minimum coolant temperature and check for leaks.

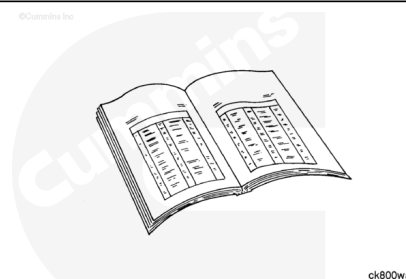


QSK60 Marine Applications

The push rods **must** be seated correctly in the cam followers.

The injector push rods are thicker in the middle. The valve push rods are the same for both intake and exhaust.

- Lubricate the sockets in the cam followers with clean engine oil.
- Install the push rods. Refer to Procedure 004-014 in Section 4 (/qs3/pubsys2/xml/en/procedures/56/56-004-014-tr.html).
- Install the crossheads. Refer to Procedure 002-001 in Section 2 (/qs3/pubsys2/xml/en/procedures/56/56-002-001-tr.html).
- Install the rocker lever assemblies. Refer to Procedure 003-009 in Section 3

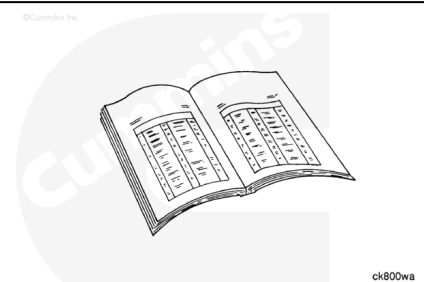


(/qs3/pubsys2/xml/en/procedures/56/56-003-009-tr.html).

- Adjust the rocker levers. Refer to Procedure 003-006 in Section 3 (/qs3/pubsys2/xml/en/procedures/56/56-003-006-tr.html).
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3 (/qs3/pubsys2/xml/en/procedures/56/56-003-011-tr.html).
- Install the aftercooler heat shields. Refer to Procedure 010-129 in Section 10 (/qs3/pubsys2/xml/en/procedures/56/56-010-129-tr.html).
- Fill the cooling system. Refer to Procedure 008-018 in Section 8 (/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html).
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with Electronically Actuated Injector

- Install the injectors. Refer to Procedure 006-026 in Section 6 (/qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html).
- Install the push rods. Refer to Procedure 004-014 in Section 4 (/qs3/pubsys2/xml/en/procedures/56/56-004-014-tr.html).
- Install the crossheads. Refer to Procedure 002-001 in Section 2 (/qs3/pubsys2/xml/en/procedures/56/56-002-001-tr.html).
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- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3 (/qs3/pubsys2/xml/en/procedures/56/56-003-011-tr.html).
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Last Modified: 10-Feb-2012
